



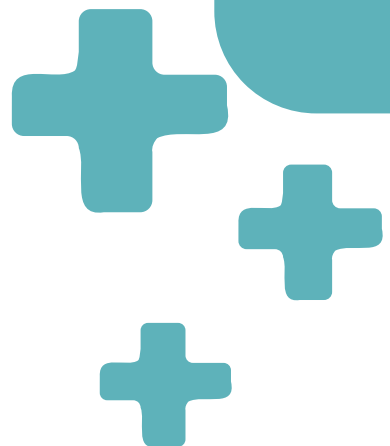
HEALTH PROVIDER FRAUD DETECTION

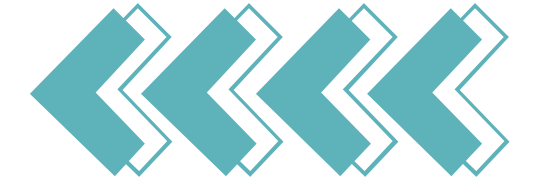




PROBLEM OVERVIEW

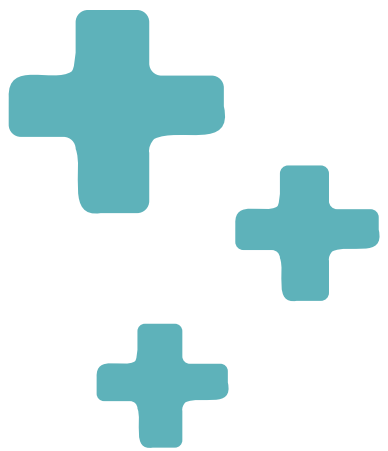
- Healthcare fraud costs the U.S. healthcare system billions annually
- Fraudulent providers may:
- Bill for services never performed
- Upcode services
- Submit claims for deceased patients
- Goal: Build a machine learning model to flag high-risk providers





WHY DOES IT MATTER?

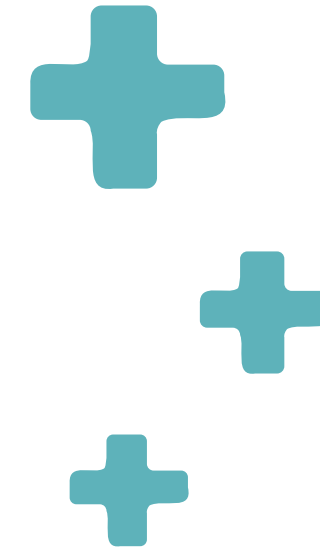
- Fraud detection reduces large financial losses
- Supports investigators by prioritizing suspicious providers
- Improves Medicare program integrity
- Manual fraud detection is slow → ML speeds up investigations



DATA USAGE

- Beneficiary Data: age, chronic conditions, date of death
- Inpatient Claims: admission dates, reimbursement amounts
- Outpatient Claims: outpatient visit and service data
- Provider Fraud Labels: identifies fraudulent vs. non-fraudulent providers

METHODOLOGY



Raw Data → Cleaning → Feature Engineering
→ Provider-Level Aggregation → Train/Val/Test Split →
Modeling → Evaluation → Error Analysis



FEATURE ENGINEERING

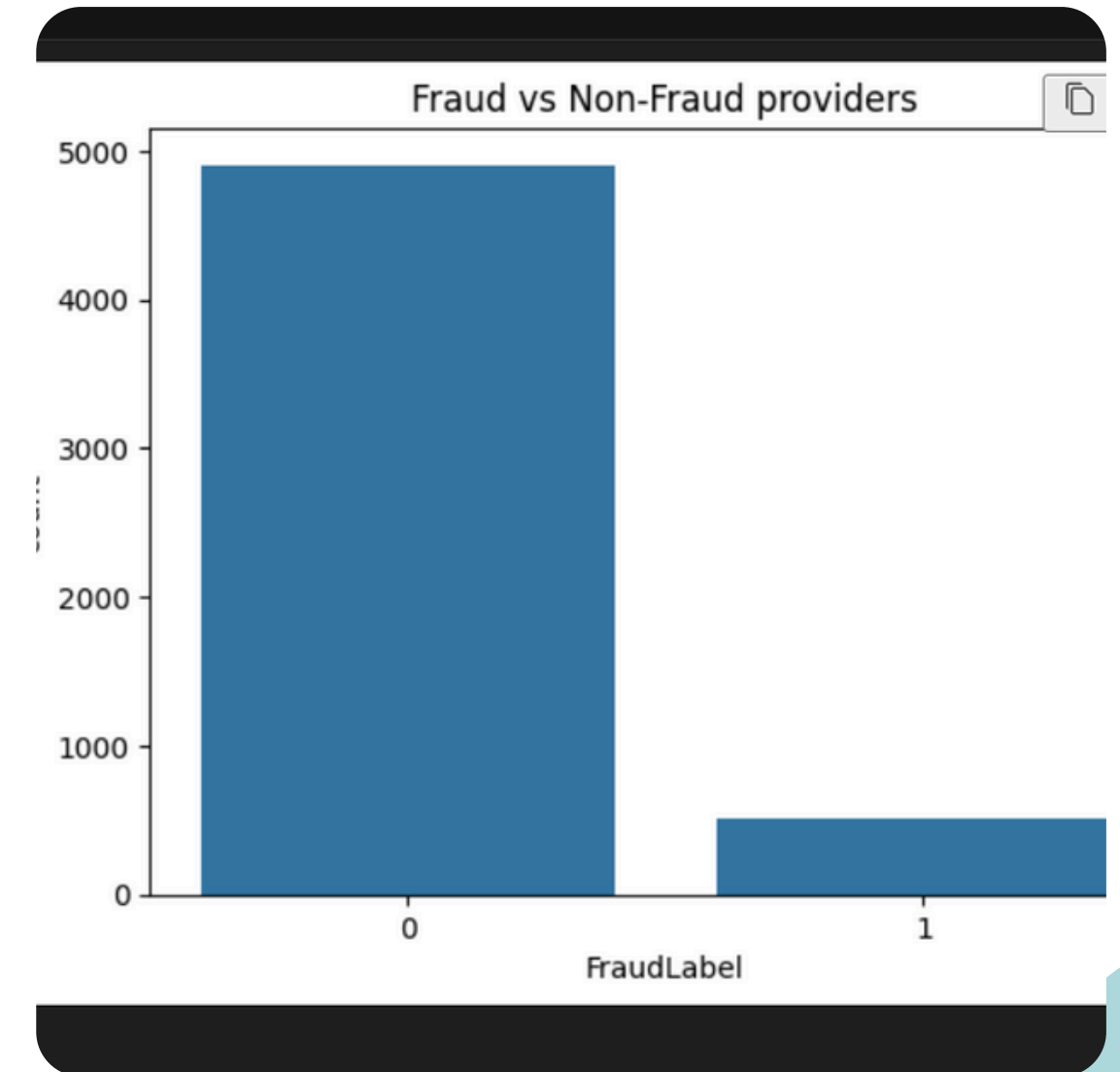
- Number of claims per provider
- Total reimbursed amount
- Average claim duration
- Number of diagnosis codes
- Number of unique beneficiaries
- Beneficiary age distribution
- Percentage of deceased beneficiaries
- Inpatient/Outpatient claim ratios
- Reimbursement per claim
- Why: These features represent provider “behavior patterns”



HANDLING CLASS IMBALANCE



- Fraud cases represent ~8–10% of providers
- Imbalance handling methods used:
- `class_weight="balanced"`
- SMOTE or oversampling
- Focused on PR-AUC (better for rare-fraud detection)



MODELS TESTED

- **Logistic Regression (baseline)**
- **Decision Tree**
- **Random Forest**
- **Gradient Boosting (best performance)**
- **SVM (optional)**



COMPARASION

Compared models using:

- PR-AUC
- ROC-AUC
- Precision, Recall, F



```
metrics_df = pd.DataFrame(summary_rows).sort_values("pr_auc", ascending=False)
metrics_df
```

Python

model	precision	recall	f1	roc_auc	pr_auc
GradientBoosting	0.797101	0.544554	0.647059	0.955895	0.772946
RandomForest	0.606557	0.732673	0.663677	0.950934	0.764216
LogReg	0.418269	0.861386	0.563107	0.946821	0.740478
DecisionTree	0.351779	0.881188	0.502825	0.876101	0.689820
SVM	0.373444	0.891089	0.526316	0.924072	0.438150

```
# Cell 12: Retrain best model on Train+Validation and save both the model and the metrics
```

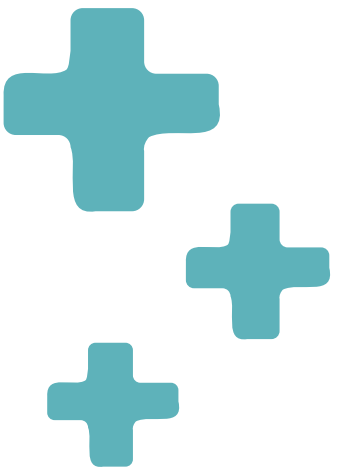
```
# 1) Combine train and validation sets
```

```
X_train_full = pd.concat([X_train, X_val], axis=0)
```

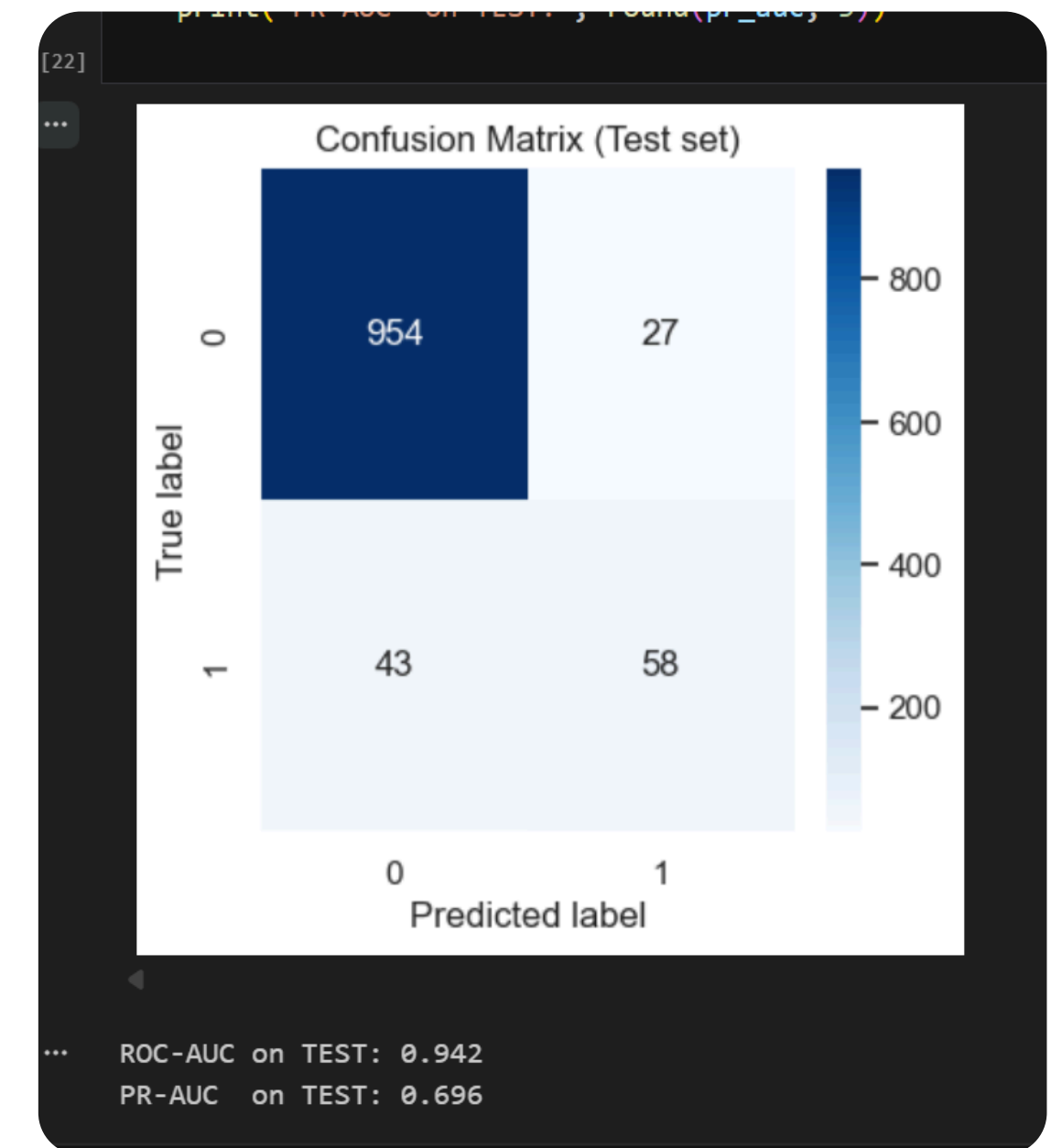
```
y_train_full = pd.concat([y_train, y_val], axis=0)
```

```
X_train_full, X_val_full, y_train_full, y_val_full = train_test_split(X_train_full, y_train_full, test_size=0.2, random_state=42)
```

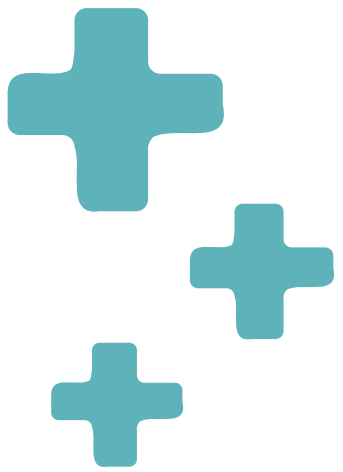
FINAL MODEL PERFORMANCE



- Final model evaluated on unseen test data
- Key metrics:
- Precision
- Recall
- F1-Score
- ROC-AUC
- PR-AUC



ROC



- Shows trade-off between True Positive Rate and False Positive Rate
- Higher AUC (> 0.85) indicates strong separation ability

```
Python  
Number of False Positives: 27  
Number of False Negatives: 43
```

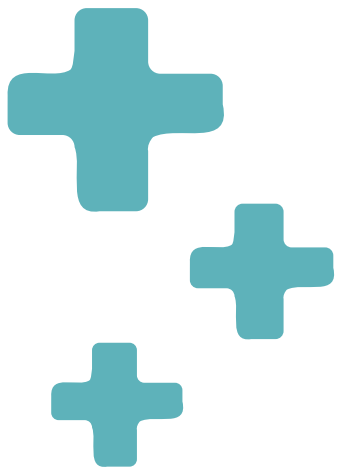
```
=== Top 3 False Positives (legitimate providers flagged as fraud) ===
```

	Provider	FraudLabel	pred_label	pred_proba	total_claim_count	total
782	PRV55916	0	1	0.970963	127.0	
59	PRV51456	0	1	0.932717	135.0	
156	PRV52063	0	1	0.918394	792.0	

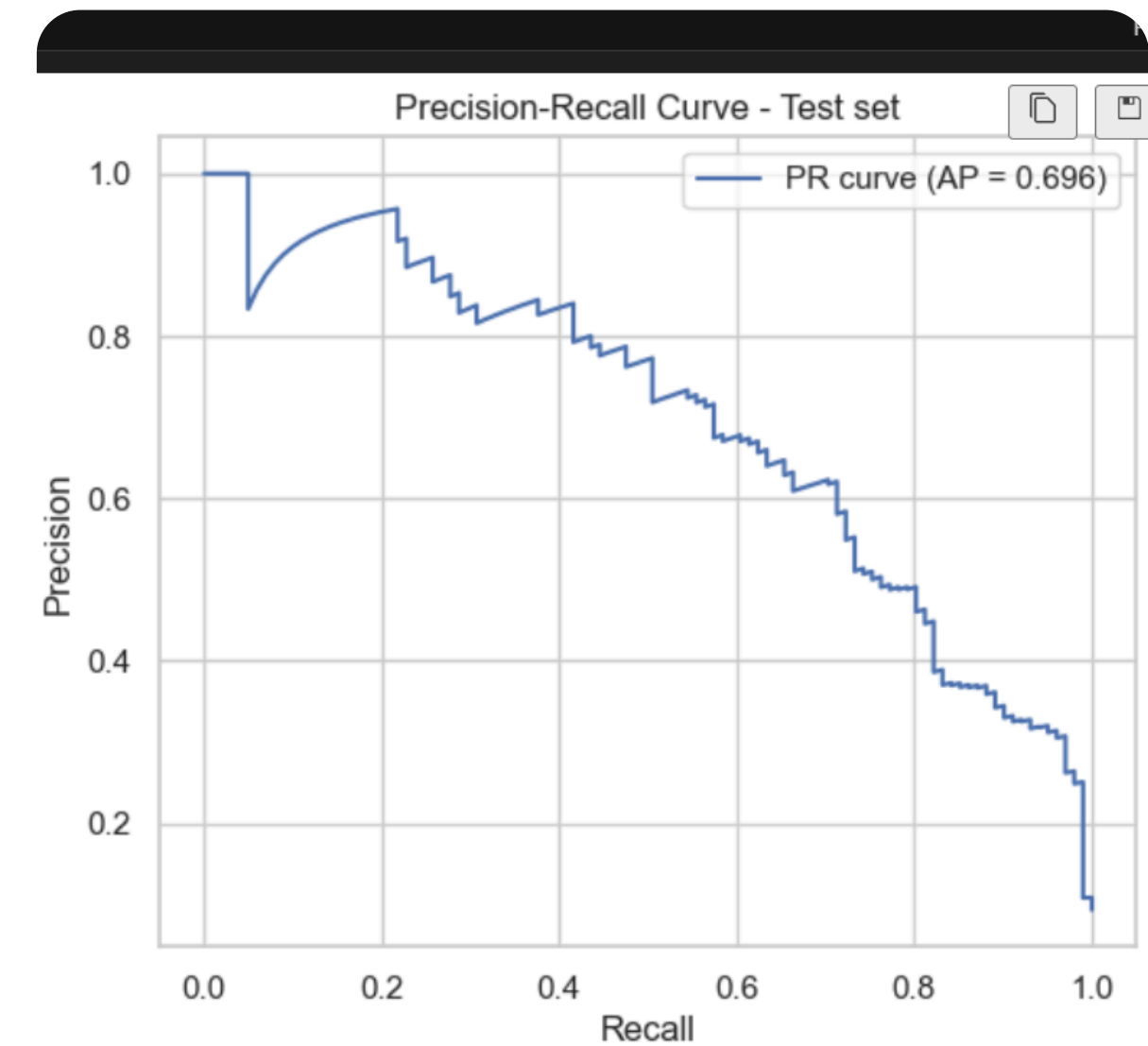
```
=== Top 3 False Negatives (fraudulent providers missed) ===
```

	Provider	FraudLabel	pred_label	pred_proba	total_claim_count	total
877	PRV56523	1	0	0.005051	21.0	
769	PRV55852	1	0	0.012541	60.0	
1063	PRV57672	1	0	0.015109	425.0	

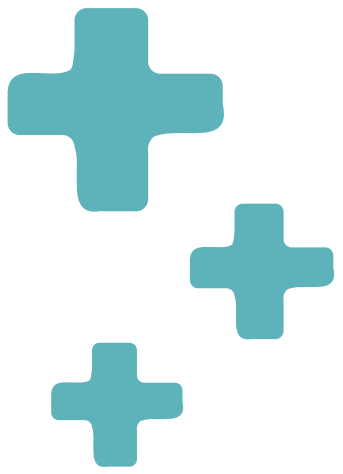
PRECISION RECALL



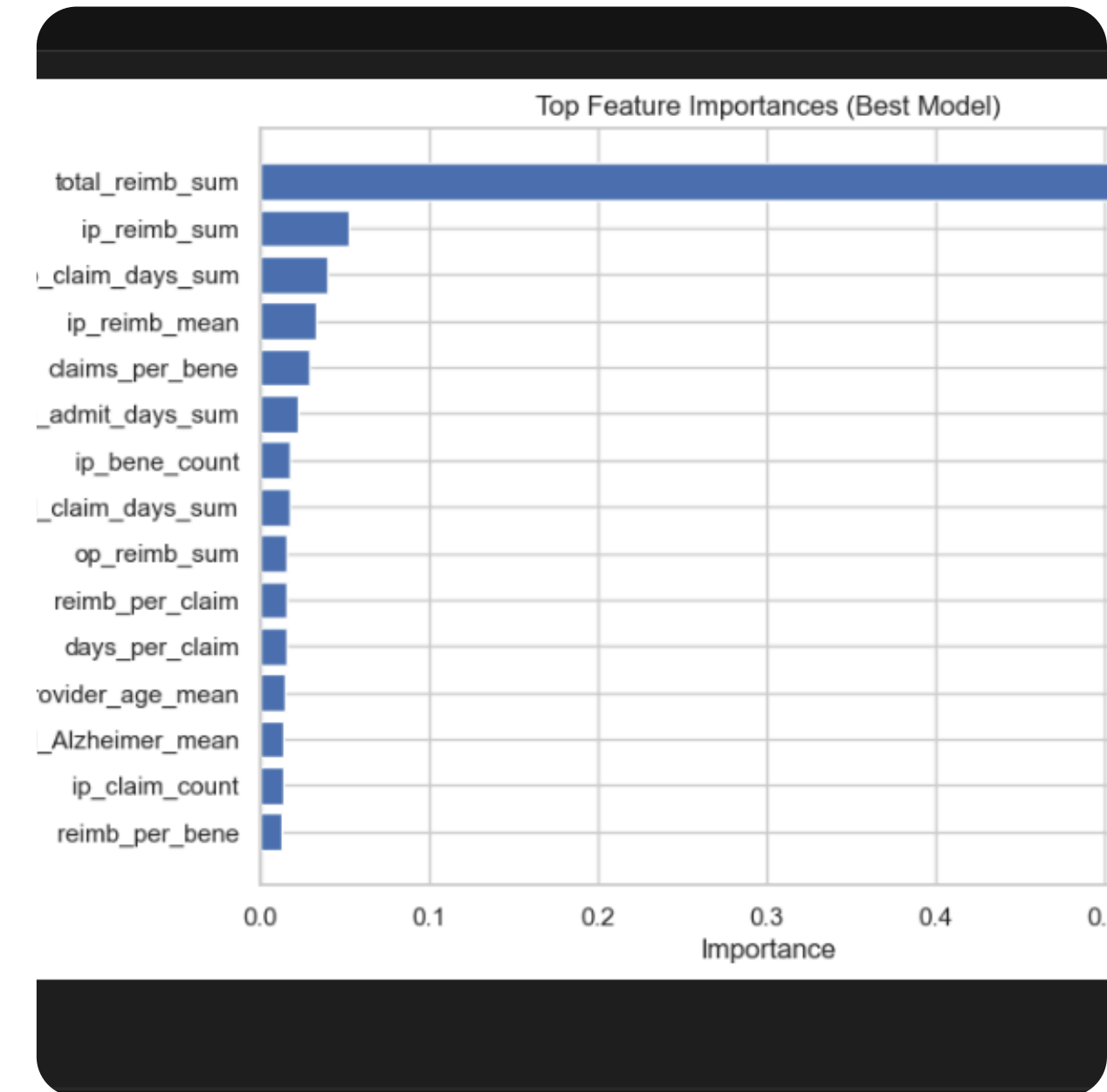
- PR curve focuses on minority fraud class
- High PR-AUC indicates reliable fraud detection



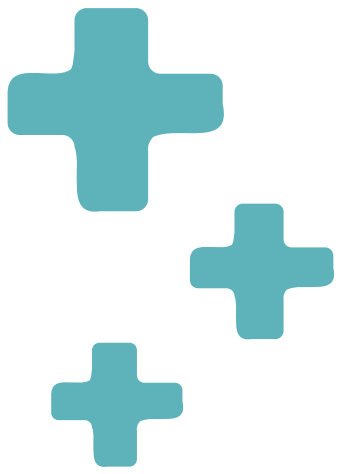
FEATURE IMPORTANCE



- Top important features include:
- Total reimbursed amount
- Claims per beneficiary
- IP/OP claim ratio
- Average claim duration

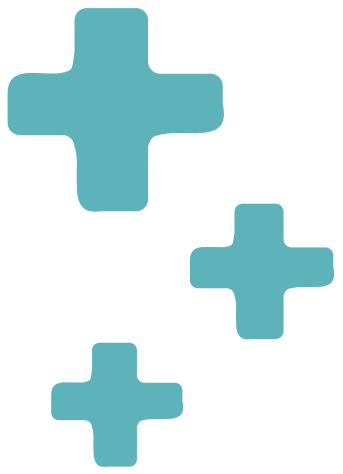


ERROR ANALYSIS



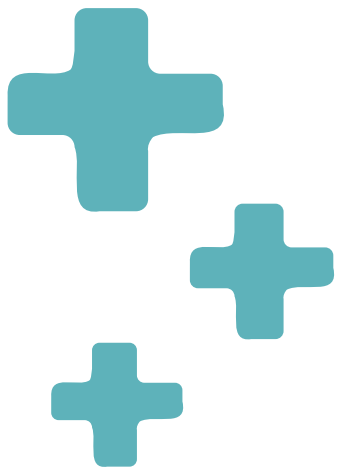
- False Positives:
- Legitimate providers incorrectly flagged
- Leads to wasted investigation resources
- False Negatives:
- Fraudulent providers missed
- Most serious → undetected financial loss
- Observed patterns:
- Some false negatives had few claims
- Some false positives had unusually long claim durations

BUSINESS IMPACT

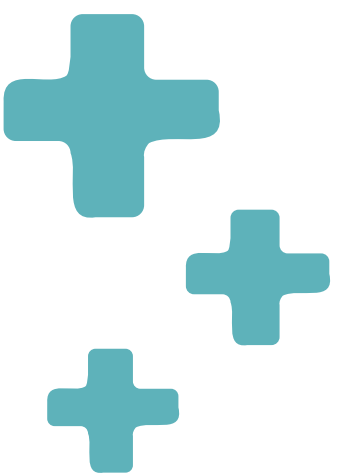


- Reduces audit workload by prioritizing risky providers
- Helps prevent financial losses
- Improves quality and integrity of Medicare operations
- Enables continuous ML-based fraud monitoring

FUTURE WORK



- Add temporal modeling (time-based fraud trends)
- Experiment with XGBoost / LightGBM
- Deploy model as an API for live scoring
- More advanced feature engineering and deep learning approach



**THANK YOU FOR YOUR
ATTENTION**

Any questions?



